

Sealed Experiment Record

Sealed 2026-10-08 · Signed by the K-Veritas server
Session b69c95c9 · Data hash f8c4...d2a5

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-08 09:12 and 09:32 UTC. Reported results: train_loss 0.999387. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	b69c95c9-99dc-4171-a23e-ad6a2d85d581
Initialized	2026-10-08T09:12:41Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	0ce6488d2430bab9170d2520e1236111407267647f6a7801a9b8acca910d62b1
Disclosure	open
Source code, run 1	54787851718d0484f8bb5dfc011f3636b1b87c72a554586f5e29531b02a6f13f

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	19 minutes 54 secon...	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
train_loss	0.849146	410 (41 readings)	explicit
train_runtime_s	1181		explicit
train_samples_per_second	2.769		explicit

Inline claim	Value	Phase	Output line
train_loss	0.999387		75

4 Execution evidence

Coherence

Coherent (0.52): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	1203.3 s	9.34 GB	95%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 459 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	09:12:42	9a23c0ad26669ad1	459
2	run_end	09:32:35	9a23c0ad26669ad1	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 617 files read, 0 files written, 18 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-08T09:32:38.125000+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	f8c48c4d67d199557674230f0069a6339f880f9716e54532e583e14e3bbfd2a5
Code bundle hash	0a680f8869eee4c0bcad32ae50d68487a9b387ee0a0bd47537b0f6c8a9da5586

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
train_loss	2.00637	10	32	explicit
train_loss	1.89885	20	33	explicit
train_loss	1.4289	30	34	explicit
train_loss	1.1568	40	35	explicit
train_loss	1.13845	50	36	explicit
train_loss	1.02476	60	37	explicit
train_loss	1.01899	70	38	explicit
train_loss	0.923368	80	39	explicit
train_loss	1.06379	90	40	explicit
train_loss	0.948886	100	41	explicit
train_loss	1.06622	110	42	explicit
train_loss	1.02933	120	43	explicit
train_loss	1.05573	130	44	explicit
train_loss	0.924057	140	45	explicit
train_loss	0.987615	150	46	explicit
train_loss	1.01598	160	47	explicit
train_loss	0.958797	170	48	explicit
train_loss	0.940118	180	49	explicit
train_loss	0.914329	190	50	explicit
train_loss	0.939907	200	51	explicit
train_loss	0.883538	210	52	explicit
train_loss	0.863418	220	53	explicit
train_loss	0.994634	230	54	explicit
train_loss	1.03944	240	55	explicit
train_loss	0.919045	250	56	explicit
train_loss	0.876972	260	57	explicit
train_loss	0.936222	270	58	explicit
train_loss	1.04292	280	59	explicit
train_loss	0.91619	290	60	explicit
train_loss	0.805232	300	61	explicit
train_loss	0.894268	310	62	explicit
train_loss	0.796796	320	63	explicit
train_loss	0.824825	330	64	explicit
train_loss	0.853621	340	65	explicit
train_loss	0.930171	350	66	explicit
train_loss	0.789536	360	67	explicit
train_loss	0.874749	370	68	explicit
train_loss	0.839612	380	69	explicit
train_loss	0.832281	390	70	explicit
train_loss	0.857568	400	71	explicit
train_loss	0.849146	410	72	explicit
train_runtime_s	1181		73	explicit
train_samples_per_second	2.769		74	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	1	09:12:42	1.5 s	0.46 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	09:12:42	9a23c0ad26669ad1	459
1	phase train	09:12:42	9a23c0ad26669ad1	0
2	run_end	09:32:35	9a23c0ad26669ad1	0

Event	Path or command	Hash
read	kv_run.sh	
read	src/llamafactory/__init__.py	
read	src/llamafactory/extras/__init__.py	
read	src/llamafactory/extras/env.py	
read	src/llamafactory/cli.py	
read	src/llamafactory/extras/misc.py	
read	.github/ISSUE_TEMPLATE/1-bug-report.yml	
read	.github/ISSUE_TEMPLATE/2-feature-request.yml	
read	.github/ISSUE_TEMPLATE/config.yml	
read	.github/workflows/docker.yml	
read	.github/workflows/docker_npu.yml	
read	.github/workflows/docs.yml	
read	.github/workflows/label_issue.yml	
read	.github/workflows/publish.yml	
read	.github/workflows/tests.yml	
read	.github/workflows/tests_cuda.yml	
read	.github/workflows/tests_npu.yml	
read	.pre-commit-config.yaml	
read	data/alpaca_en_demo.json	
read	data/alpaca_zh_demo.json	
read	data/dataset_info.json	
read	data/dpo_en_demo.json	
read	data/dpo_zh_demo.json	
read	data/glaive_toolcall_en_demo.json	
read	data/glaive_toolcall_zh_demo.json	
read	data/identity.json	
read	data/kto_en_demo.json	
read	data/mllm_audio_demo.json	
read	data/mllm_demo.json	
read	data/mllm_video_audio_demo.json	
read	data/mllm_video_demo.json	
read	data/v1_dpo_demo.yaml	
read	data/v1_multimodal_demo.yaml	
read	data/v1_sft_demo.yaml	
read	docker/docker-cuda/docker-compose.yml	
read	docker/docker-npu/docker-compose.yml	
read	docker/docker-rocm/docker-compose.yml	
read	docker/docker-xpu/docker-compose.yml	

Event	Path or command	Hash
read	docs/_static/js/switcher.js	
read	docs/conf.py	
read	docs/en/conf.py	
read	docs/zh/conf.py	
read	examples/accelerate/fsdp2_config.yaml	
read	examples/accelerate/fsdp2_config_qwen35.yaml	
read	examples/accelerate/fsdp2_config_qwen35_moe.yaml	
read	examples/accelerate/fsdp_config.yaml	
read	examples/accelerate/fsdp_config_multiple_nodes.yaml	
read	examples/accelerate/fsdp_config_offload.yaml	
read	examples/ascend/qwen3_5_full_sft_fsdp2.yaml	
read	examples/ascend/qwen3_5moe_lora_sft_fsdp2.yaml	
read	examples/ascend/qwen3_full_sft_fsdp2.yaml	
read	examples/ascend/qwen3moe_full_sft_fsdp2.yaml	
read	examples/ascend/qwen3v1moe_full_sft_fsdp2.yaml	
read	examples/ascend/qwen3v1moe_lora_sft_fsdp2.yaml	
read	examples/deepspeed/ds_z0_config.json	
read	examples/deepspeed/ds_z2_autotp_config.json	
read	examples/deepspeed/ds_z2_config.json	
read	examples/deepspeed/ds_z2_offload_config.json	
read	examples/deepspeed/ds_z3_config.json	
read	examples/deepspeed/ds_z3_fp8_config.json	
read	examples/deepspeed/ds_z3_offload_config.json	
read	examples/extras/adam_mini/qwen2_full_sft.yaml	
read	examples/extras/apollo/llama3_full_sft.yaml	
read	examples/extras/asft/llama2_full_asft.yaml	
read	examples/extras/asft/qwen2_full_asft.yaml	
read	examples/extras/badam/llama3_full_sft.yaml	
read	examples/extras/dft/qwen2_full_sft.yaml	
read	examples/extras/eaft/qwen25_05b_eaft_full.yaml	
read	examples/extras/fp8/llama3_fp8_deepspeed_sft.yaml	
read	examples/extras/fp8/llama3_fp8_fsdp_sft.yaml	
read	examples/extras/fsdp_qlora/llama3_lora_sft.yaml	
read	examples/extras/fsdp_qlora/train.sh	
read	examples/extras/galore/llama3_full_sft.yaml	
read	examples/extras/llama_pro/expand.sh	
read	examples/extras/llama_pro/llama3_freeze_sft.yaml	
read	examples/extras/loraplus/llama3_lora_sft.yaml	
read	examples/extras/mod/llama3_full_sft.yaml	
read	examples/extras/multi_tokens/tokens_cfg.yaml	
read	examples/extras/muon/qwen2_full_sft.yaml	
read	examples/extras/nlg_eval/llama3_lora_predict.yaml	
read	examples/extras/oft/llama3_ofsft.yaml	
read	examples/extras/oft/qwen2_5v1_ofsft.yaml	
read	examples/extras/pissa/init.sh	
read	examples/extras/pissa/llama3_lora_sft.yaml	
read	examples/extras/qoft/llama3_ofsft_awq.yaml	
read	examples/extras/qoft/llama3_ofsft_bnb_npu.yaml	
read	examples/extras/qoft/llama3_ofsft_gptq.yaml	
read	examples/inference/mossv1.yaml	

Event	Path or command	Hash
read	examples/inference/qwen3.yaml	
read	examples/inference/qwen3_full_sft.yaml	
read	examples/inference/qwen3_lora_sft.yaml	
read	examples/inference/qwen3vl.yaml	
read	examples/ktransformers/accelerate/fsdp2_kt_bf16.yaml	
read	examples/ktransformers/accelerate/fsdp2_kt_int4.yaml	
read	examples/ktransformers/accelerate/fsdp2_kt_int8.yaml	
read	examples/ktransformers/accelerate/fsdp2_kt_int8_1gpu.yaml	
read	examples/ktransformers/accelerate/fsdp2_kt_int8_8gpu.yaml	
read	examples/ktransformers/train_lora/deepseek_v2_lora_sft_kt.yaml	
read	examples/ktransformers/train_lora/deepseek_v3_int8_lora_sft_kt.yaml	
read	examples/ktransformers/train_lora/deepseek_v3_lora_sft_kt.yaml	
read	examples/ktransformers/train_lora/qwen3_5moe_lora_sft_kt.yaml	
read	examples/ktransformers/train_lora/qwen3moe_lora_sft_kt.yaml	
read	examples/ktransformers/train_lora/qwen3vlmoe_lora_sft_kt.yaml	
read	examples/megatron/qwen2_vl_full.yaml	
read	examples/megatron/qwen3_moe_full.yaml	
read	examples/megatron_bridge/llama3_sft.yaml	
read	examples/merge_lora/moss_vl_lora_sft.yaml	
read	examples/merge_lora/qwen3_full_sft.yaml	
read	examples/merge_lora/qwen3_gptq.yaml	
read	examples/merge_lora/qwen3_lora_sft.yaml	
read	examples/merge_lora/qwen3vl_lora_sft.yaml	
read	examples/train_full/moss_vl_full_sft.yaml	
read	examples/train_full/qwen3_full_sft.yaml	
read	examples/train_full/qwen3vl_full_sft.yaml	
read	examples/train_lora/moss_vl_lora_sft.yaml	
read	examples/train_lora/qwen3_lora_dpo.yaml	
read	examples/train_lora/qwen3_lora_kto.yaml	
read	examples/train_lora/qwen3_lora_pretrain.yaml	
read	examples/train_lora/qwen3_lora_reward.yaml	
read	examples/train_lora/qwen3_lora_sft.sh	
read	examples/train_lora/qwen3_lora_sft.yaml	
read	examples/train_lora/qwen3_lora_sft_ds3.yaml	
read	examples/train_lora/qwen3_lora_sft_ray.yaml	
read	examples/train_lora/qwen3_preprocess.yaml	
read	examples/train_lora/qwen3vl_lora_dpo.yaml	
read	examples/train_lora/qwen3vl_lora_sft.yaml	
read	examples/train_qlora/llama3_lora_sft_aqlm.yaml	
read	examples/train_qlora/llama3_lora_sft_awq.yaml	
read	examples/train_qlora/llama3_lora_sft_gptq.yaml	
read	examples/train_qlora/qwen3_lora_sft_bnb_npu.yaml	
read	examples/train_qlora/qwen3_lora_sft_otfq.yaml	
read	examples/v1/train_batching_strategy/train_full_fsdp2_batching_normal.yaml	
read	examples/v1/train_batching_strategy/train_full_fsdp2_dynamic_batching.yaml	
read	examples/v1/train_batching_strategy/train_full_fsdp2_dynamic_padding_free...	
read	examples/v1/train_batching_strategy/train_full_fsdp2_padding_free.yaml	
read	examples/v1/train_freeze/train_freeze_sft.yaml	
read	examples/v1/train_full/train_full_chunk_loss.yaml	
read	examples/v1/train_full/train_full_deepspeed.yaml	

Event	Path or command	Hash
read	examples/v1/train_full/train_full_fsd2.yaml	
read	examples/v1/train_full/train_full_liger_kernel.yaml	
read	examples/v1/train_full/train_full_multimodal_ulysses_cp.yaml	
read	examples/v1/train_full/train_full_muon.yaml	
read	examples/v1/train_full/train_full_qwen3_moe_fsdpturbo_ep_fsd2.yaml	
read	examples/v1/train_full/train_full_ulysses_cp.yaml	
read	examples/v1/train_full/train_multimodal.yaml	
read	examples/v1/train_lora/export_lora.yaml	
read	examples/v1/train_lora/train_lora_dpo.yaml	
read	examples/v1/train_lora/train_lora_sft.yaml	
read	examples/v1/train_lora/train_lora_sft_rank0.yaml	
read	examples/v1/train_qlora/quantization.yaml	
read	kv_qwen3_lora_sft.yaml	
read	pyproject.toml	
read	scripts/api_example/test_image.py	
read	scripts/api_example/test_toolcall.py	
read	scripts/bench_qwen.py	
read	scripts/convert_ckpt/llamafy_baichuan2.py	
read	scripts/convert_ckpt/llamafy_qwen.py	
read	scripts/convert_ckpt/tiny_llama4.py	
read	scripts/convert_ckpt/tiny_qwen3.py	
read	scripts/dcp2hf.py	
read	scripts/eval_bleu_rouge.py	
read	scripts/hf2dcp.py	
read	scripts/llama_pro.py	
read	scripts/loftq_init.py	
read	scripts/megatron_merge.py	
read	scripts/pissa_init.py	
read	scripts/qwen_omni_merge.py	
read	scripts/stat_utils/cal_flops.py	
read	scripts/stat_utils/cal_lr.py	
read	scripts/stat_utils/cal_mfu.py	
read	scripts/stat_utils/cal_ppl.py	
read	scripts/stat_utils/length_cdf.py	
read	scripts/vllm_infer.py	
read	src/api.py	
read	src/llamafactory/api/__init__.py	
read	src/llamafactory/api/app.py	
read	src/llamafactory/api/chat.py	
read	src/llamafactory/api/common.py	
read	src/llamafactory/api/protocol.py	
read	src/llamafactory/chat/__init__.py	
read	src/llamafactory/chat/base_engine.py	
read	src/llamafactory/chat/chat_model.py	
read	src/llamafactory/chat/hf_engine.py	
read	src/llamafactory/chat/sglang_engine.py	
read	src/llamafactory/chat/vllm_engine.py	
read	src/llamafactory/data/__init__.py	
read	src/llamafactory/data/collator.py	
read	src/llamafactory/data/converter.py	

Event	Path or command	Hash
read	src/llamafactory/data/data_utils.py	
read	src/llamafactory/data/formatter.py	
read	src/llamafactory/data/loader.py	
read	src/llamafactory/data/mm_plugin.py	
read	src/llamafactory/data/parser.py	
read	src/llamafactory/data/processor/__init__.py	
read	src/llamafactory/data/processor/feedback.py	
read	src/llamafactory/data/processor/pairwise.py	
read	src/llamafactory/data/processor/pretrain.py	
read	src/llamafactory/data/processor/processor_utils.py	
read	src/llamafactory/data/processor/supervised.py	
read	src/llamafactory/data/processor/unsupervised.py	
read	src/llamafactory/data/template.py	
read	src/llamafactory/data/tool_utils.py	
read	src/llamafactory/eval/__init__.py	
read	src/llamafactory/eval/evaluator.py	
read	src/llamafactory/eval/template.py	
read	src/llamafactory/extras/constants.py	
read	src/llamafactory/extras/logging.py	
read	src/llamafactory/extras/packages.py	
read	src/llamafactory/extras/ploting.py	
read	src/llamafactory/hparams/__init__.py	
read	src/llamafactory/hparams/data_args.py	
read	src/llamafactory/hparams/evaluation_args.py	
read	src/llamafactory/hparams/finetuning_args.py	
read	src/llamafactory/hparams/generating_args.py	
read	src/llamafactory/hparams/megatron_bridge_args.py	
read	src/llamafactory/hparams/model_args.py	
read	src/llamafactory/hparams/parser.py	
read	src/llamafactory/hparams/training_args.py	
read	src/llamafactory/launcher.py	
read	src/llamafactory/model/__init__.py	
read	src/llamafactory/model/adapter.py	
read	src/llamafactory/model/loader.py	
read	src/llamafactory/model/model_utils/__init__.py	
read	src/llamafactory/model/model_utils/attention.py	
read	src/llamafactory/model/model_utils/checkpointing.py	
read	src/llamafactory/model/model_utils/embedding.py	
read	src/llamafactory/model/model_utils/kv_cache.py	
read	src/llamafactory/model/model_utils/liger_kernel.py	
read	src/llamafactory/model/model_utils/longlora.py	
read	src/llamafactory/model/model_utils/misc.py	
read	src/llamafactory/model/model_utils/mod.py	
read	src/llamafactory/model/model_utils/moe.py	
read	src/llamafactory/model/model_utils/packing.py	
read	src/llamafactory/model/model_utils/quantization.py	
read	src/llamafactory/model/model_utils/rope.py	
read	src/llamafactory/model/model_utils/unsloth.py	
read	src/llamafactory/model/model_utils/valuehead.py	
read	src/llamafactory/model/model_utils/visual.py	

Event	Path or command	Hash
read	src/llamafactory/model/patcher.py	
read	src/llamafactory/third_party/__init__.py	
read	src/llamafactory/third_party/muon/__init__.py	
read	src/llamafactory/third_party/muon/muon.py	
read	src/llamafactory/third_party/triton/chunk_delta_h.py	
read	src/llamafactory/third_party/triton/chunk_gated_delta_rule.py	
read	src/llamafactory/third_party/triton/chunk_o.py	
read	src/llamafactory/third_party/triton/chunk_scaled_dot_kkt.py	
read	src/llamafactory/third_party/triton/cumsum.py	
read	src/llamafactory/third_party/triton/solve_tril.py	
read	src/llamafactory/third_party/triton/utils.py	
read	src/llamafactory/third_party/triton/wy_fast.py	
read	src/llamafactory/train/__init__.py	
read	src/llamafactory/train/callbacks.py	
read	src/llamafactory/train/dpo/__init__.py	
read	src/llamafactory/train/dpo/trainer.py	
read	src/llamafactory/train/dpo/workflow.py	
read	src/llamafactory/train/fp8_utils.py	
read	src/llamafactory/train/hyper_parallel/__init__.py	
read	src/llamafactory/train/hyper_parallel/trainer.py	
read	src/llamafactory/train/hyper_parallel/workflow.py	
read	src/llamafactory/train/kto/__init__.py	
read	src/llamafactory/train/kto/trainer.py	
read	src/llamafactory/train/kto/workflow.py	
read	src/llamafactory/train/mca/__init__.py	
read	src/llamafactory/train/mca/trainer.py	
read	src/llamafactory/train/mca/workflow.py	
read	src/llamafactory/train/megatron_bridge/__init__.py	
read	src/llamafactory/train/megatron_bridge/config_builder.py	
read	src/llamafactory/train/megatron_bridge/dataset_export.py	
read	src/llamafactory/train/megatron_bridge/workflow.py	
read	src/llamafactory/train/ppo/__init__.py	
read	src/llamafactory/train/ppo/ppo_utils.py	
read	src/llamafactory/train/ppo/trainer.py	
read	src/llamafactory/train/ppo/workflow.py	
read	src/llamafactory/train/pt/__init__.py	
read	src/llamafactory/train/pt/trainer.py	
read	src/llamafactory/train/pt/workflow.py	
read	src/llamafactory/train/rm/__init__.py	
read	src/llamafactory/train/rm/metric.py	
read	src/llamafactory/train/rm/trainer.py	
read	src/llamafactory/train/rm/workflow.py	
read	src/llamafactory/train/sft/__init__.py	
read	src/llamafactory/train/sft/metric.py	
read	src/llamafactory/train/sft/trainer.py	
read	src/llamafactory/train/sft/workflow.py	
read	src/llamafactory/train/test_utils.py	
read	src/llamafactory/train/trainer_utils.py	
read	src/llamafactory/train/tuner.py	
read	src/llamafactory/v1/__init__.py	

Event	Path or command	Hash
read	src/llamafactory/v1/accelerator/__init__.py	
read	src/llamafactory/v1/accelerator/helper.py	
read	src/llamafactory/v1/accelerator/interface.py	
read	src/llamafactory/v1/accelerator/profiler.py	
read	src/llamafactory/v1/config/__init__.py	
read	src/llamafactory/v1/config/arg_parser.py	
read	src/llamafactory/v1/config/arg_utils.py	
read	src/llamafactory/v1/config/data_args.py	
read	src/llamafactory/v1/config/model_args.py	
read	src/llamafactory/v1/config/sample_args.py	
read	src/llamafactory/v1/config/training_args.py	
read	src/llamafactory/v1/core/__init__.py	
read	src/llamafactory/v1/core/base_sampler.py	
read	src/llamafactory/v1/core/base_trainer.py	
read	src/llamafactory/v1/core/data_engine.py	
read	src/llamafactory/v1/core/model_engine.py	
read	src/llamafactory/v1/core/rendering/__init__.py	
read	src/llamafactory/v1/core/rendering/escape.py	
read	src/llamafactory/v1/core/rendering/format.py	
read	src/llamafactory/v1/core/rendering/rendering.py	
read	src/llamafactory/v1/core/utils/__init__.py	
read	src/llamafactory/v1/core/utils/batching.py	
read	src/llamafactory/v1/core/utils/callback.py	
read	src/llamafactory/v1/core/utils/checkpoint.py	
read	src/llamafactory/v1/core/utils/collation.py	
read	src/llamafactory/v1/core/utils/inference_engine.py	
read	src/llamafactory/v1/launcher.py	
read	src/llamafactory/v1/plugins/__init__.py	
read	src/llamafactory/v1/plugins/data_plugins/__init__.py	
read	src/llamafactory/v1/plugins/data_plugins/converter.py	
read	src/llamafactory/v1/plugins/data_plugins/loader.py	
read	src/llamafactory/v1/plugins/model_plugins/__init__.py	
read	src/llamafactory/v1/plugins/model_plugins/add_token.py	
read	src/llamafactory/v1/plugins/model_plugins/chunk_loss.py	
read	src/llamafactory/v1/plugins/model_plugins/deepspeed_utils.py	
read	src/llamafactory/v1/plugins/model_plugins/initialization.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/__init__.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/base.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/interface.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/liger_kernel_ops.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/__init__.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/linear_attention/fla.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/mlp/__init__.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/mlp/cuda_fused_moe....	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/mlp/npu_fused_moe.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/mlp/npu_swiglu.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/mlp/triton_grouped_g...	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/rms_norm/__init__.py	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/rms_norm/npu_rms_n...	
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/rope/__init__.py	

Event	Path or command	Hash
read	src/llamafactory/v1/plugins/model_plugins/kernels/ops/rope/npu_rope.py	
read	src/llamafactory/v1/plugins/model_plugins/parallelization/batch.py	
read	src/llamafactory/v1/plugins/model_plugins/parallelization/gdn_attention.py	
read	src/llamafactory/v1/plugins/model_plugins/parallelization/hook.py	
read	src/llamafactory/v1/plugins/model_plugins/parallelization/seq_comm.py	
read	src/llamafactory/v1/plugins/model_plugins/parallelization/sequence_parallel.py	
read	src/llamafactory/v1/plugins/model_plugins/parallelization/ulysses.py	
read	src/llamafactory/v1/plugins/model_plugins/peft.py	
read	src/llamafactory/v1/plugins/model_plugins/quantization.py	
read	src/llamafactory/v1/plugins/sampler_plugins/__init__.py	
read	src/llamafactory/v1/plugins/sampler_plugins/vllm.py	
read	src/llamafactory/v1/plugins/trainer_plugins/__init__.py	
read	src/llamafactory/v1/plugins/trainer_plugins/batching.py	
read	src/llamafactory/v1/plugins/trainer_plugins/distributed/__init__.py	
read	src/llamafactory/v1/plugins/trainer_plugins/distributed/base.py	
read	src/llamafactory/v1/plugins/trainer_plugins/distributed/deepspeed.py	
read	src/llamafactory/v1/plugins/trainer_plugins/distributed/fsdp2.py	
read	src/llamafactory/v1/plugins/trainer_plugins/distributed/fsdp_turbo.py	
read	src/llamafactory/v1/plugins/trainer_plugins/distributed/interface.py	
read	src/llamafactory/v1/plugins/trainer_plugins/lr_scheduler.py	
read	src/llamafactory/v1/plugins/trainer_plugins/optimizers/__init__.py	
read	src/llamafactory/v1/plugins/trainer_plugins/optimizers/muon_optimizer.py	
read	src/llamafactory/v1/plugins/trainer_plugins/optimizers/optimizer.py	
read	src/llamafactory/v1/samplers/cli_sampler.py	
read	src/llamafactory/v1/trainers/__init__.py	
read	src/llamafactory/v1/trainers/dpo_trainer.py	
read	src/llamafactory/v1/trainers/rm_trainer.py	
read	src/llamafactory/v1/trainers/sft_trainer.py	
read	src/llamafactory/v1/utils/__init__.py	
read	src/llamafactory/v1/utils/callbacks/__init__.py	
read	src/llamafactory/v1/utils/callbacks/logging_callback.py	
read	src/llamafactory/v1/utils/callbacks/trainer_callback.py	
read	src/llamafactory/v1/utils/constants.py	
read	src/llamafactory/v1/utils/dtype.py	
read	src/llamafactory/v1/utils/env.py	
read	src/llamafactory/v1/utils/helper.py	
read	src/llamafactory/v1/utils/logging.py	
read	src/llamafactory/v1/utils/objects.py	
read	src/llamafactory/v1/utils/packages.py	
read	src/llamafactory/v1/utils/plugin.py	
read	src/llamafactory/v1/utils/pytest.py	
read	src/llamafactory/v1/utils/types.py	
read	src/llamafactory/webui/__init__.py	
read	src/llamafactory/webui/chatter.py	
read	src/llamafactory/webui/common.py	
read	src/llamafactory/webui/components/__init__.py	
read	src/llamafactory/webui/components/chatbot.py	
read	src/llamafactory/webui/components/data.py	
read	src/llamafactory/webui/components/eval.py	
read	src/llamafactory/webui/components/export.py	

Event	Path or command	Hash
read	src/llamafactory/webui/components/footer.py	
read	src/llamafactory/webui/components/infer.py	
read	src/llamafactory/webui/components/top.py	
read	src/llamafactory/webui/components/train.py	
read	src/llamafactory/webui/control.py	
read	src/llamafactory/webui/css.py	
read	src/llamafactory/webui/engine.py	
read	src/llamafactory/webui/interface.py	
read	src/llamafactory/webui/locales.py	
read	src/llamafactory/webui/manager.py	
read	src/llamafactory/webui/runner.py	
read	src/train.py	
read	src/webui.py	
read	tests/check_license.py	
read	tests/conftest.py	
read	tests/data/processor/test_feedback.py	
read	tests/data/processor/test_pairwise.py	
read	tests/data/processor/test_processor_utils.py	
read	tests/data/processor/test_supervised.py	
read	tests/data/processor/test_unsupervised.py	
read	tests/data/test_collator.py	
read	tests/data/test_converter.py	
read	tests/data/test_formatter.py	
read	tests/data/test_loader.py	
read	tests/data/test_mm_plugin.py	
read	tests/data/test_moss_vl_plugin.py	
read	tests/data/test_moss_vl_training_configs.py	
read	tests/data/test_template.py	
read	tests/e2e/test_chat.py	
read	tests/e2e/test_sglang.py	
read	tests/e2e/test_train.py	
read	tests/eval/test_eval_template.py	
read	tests/model/model_utils/test_add_tokens.py	
read	tests/model/model_utils/test_attention.py	
read	tests/model/model_utils/test_checkpointing.py	
read	tests/model/model_utils/test_embedding.py	
read	tests/model/model_utils/test_misc.py	
read	tests/model/model_utils/test_packing.py	
read	tests/model/model_utils/test_visual.py	
read	tests/model/test_base.py	
read	tests/model/test_freeze.py	
read	tests/model/test_full.py	
read	tests/model/test_lora.py	
read	tests/model/test_pissa.py	
read	tests/train/megatron_bridge/conftest.py	
read	tests/train/megatron_bridge/test_config_builder.py	
read	tests/train/megatron_bridge/test_dataset_export.py	
read	tests/train/megatron_bridge/test_megatron_bridge_gpu.py	
read	tests/train/megatron_bridge/test_model_support.py	
read	tests/train/test_sft_trainer.py	

Event	Path or command	Hash
read	tests_v1/accelerator/test_interface.py	
read	tests_v1/config/test_args_parser.py	
read	tests_v1/conftest.py	
read	tests_v1/core/rendering/test_rendering.py	
read	tests_v1/core/test_data_engine.py	
read	tests_v1/core/test_model_loader.py	
read	tests_v1/core/utils/test_batching.py	
read	tests_v1/plugins/data_plugins/test_converter.py	
read	tests_v1/plugins/model_plugins/test_chunk_loss.py	
read	tests_v1/plugins/model_plugins/test_init_plugin.py	
read	tests_v1/plugins/model_plugins/test_kernel_plugin.py	
read	tests_v1/plugins/model_plugins/test_peft.py	
read	tests_v1/plugins/model_plugins/test_quantization_plugin.py	
read	tests_v1/plugins/model_plugins/test_ulysses_cp.py	
read	tests_v1/plugins/trainer_plugins/distributed/test_fsdp2.py	
read	tests_v1/plugins/trainer_plugins/distributed/test_fsdp2_weight_convert.py	
read	tests_v1/plugins/trainer_plugins/distributed/test_fsdp2turbo_ep.py	
read	tests_v1/sampler/test_cli_sampler.py	
read	tests_v1/trainers/test_dpo_loss_precision.py	
read	tests_v1/trainers/test_fsdp2_dpo_trainer.py	
read	tests_v1/trainers/test_fsdp2_sft_trainer.py	
read	.ai/CLAUDE.md	
read	.gitattributes	
read	.claude/skills/llamafactory-sft/SKILL.md	
read	.claude/skills/llamafactory-v1-docs/SKILL.md	
read	.dockerignore	
read	.env.local	
read	.github/CODE_OF_CONDUCT.md	
read	.github/CONTRIBUTING.md	
read	.github/PULL_REQUEST_TEMPLATE.md	
read	.github/SECURITY.md	
read	.github/copilot-instructions.md	
read	.github/instructions-v0.md	
read	.github/instructions-v1.md	
read	.gitignore	
read	CITATION.cff	
read	LICENSE	
read	MANIFEST.in	
read	Makefile	
read	README.md	
read	README_zh.md	
read	assets/logo.png	
read	assets/sponsors/serpapi.svg	
read	assets/sponsors/warp.jpg	
read	assets/thirdparty/colab.svg	
read	assets/thirdparty/discord.svg	
read	assets/thirdparty/dsw.svg	
read	assets/thirdparty/lab4ai.svg	
read	assets/thirdparty/online.svg	
read	data/README.md	

Event	Path or command	Hash
read	data/README_zh.md	
read	data/c4_demo.jsonl	
read	data/mllm_demo_data/1.jpg	
read	data/mllm_demo_data/1.mp3	
read	data/mllm_demo_data/1.mp4	
read	data/mllm_demo_data/2.avi	
read	data/mllm_demo_data/2.jpg	
read	data/mllm_demo_data/2.wav	
read	data/mllm_demo_data/3.flac	
read	data/mllm_demo_data/3.jpg	
read	data/mllm_demo_data/3.mp4	
read	data/mllm_demo_data/4.mp3	
read	data/mllm_demo_data/4.mp4	
read	data/reason_tool_use_demo_50.jsonl	
read	data/v1_dpo_demo.jsonl	
read	data/v1_multimodal_demo.jsonl	
read	data/v1_sft_demo.jsonl	
read	data/wiki_demo.txt	
read	docker/docker-cuda/Dockerfile	
read	docker/docker-cuda/Dockerfile.base	
read	docker/docker-cuda/Dockerfile.mbridge	
read	docker/docker-cuda/Dockerfile.megatron	
read	docker/docker-cuda/README.md	
read	docker/docker-npu/Dockerfile	
read	docker/docker-npu/OVERVIEW.md	
read	docker/docker-npu/OVERVIEW.zh.md	
read	docker/docker-npu/supported_tags.md	
read	docker/docker-rocm/Dockerfile	
read	docker/docker-xpu/Dockerfile	
read	docker/docker-xpu/README.md	
read	docs/Makefile	
read	docs/_static/css/lang-switcher.css	
read	docs/_static/css/tables.css	
read	docs/en/advanced/custom-kernels/custom-kernels.md	
read	docs/en/advanced/custom-kernels/fused-operators.md	
read	docs/en/advanced/custom-kernels/triton.md	
read	docs/en/advanced/distributed/deepspeed.md	
read	docs/en/advanced/distributed/fsdp.md	
read	docs/en/advanced/distributed/fsdpturbo-ep-efsdp.md	
read	docs/en/advanced/distributed/parallel-dp-tp-ep-sp-cp.md	
read	docs/en/advanced/ktransformers.md	
read	docs/en/advanced/lora-and-quantization/lora.md	
read	docs/en/advanced/lora-and-quantization/quantization.md	
read	docs/en/data-preparation/data-processing.md	
read	docs/en/dev-guide/core/data-engine.md	
read	docs/en/dev-guide/core/model-engine.md	
read	docs/en/dev-guide/core/trainer.md	
read	docs/en/dev-guide/plugins/data-plugins.md	
read	docs/en/dev-guide/plugins/model-plugins/initialization.md	
read	docs/en/dev-guide/plugins/model-plugins/kernels.md	

Event	Path or command	Hash
read	docs/en/dev-guide/plugins/model-plugins/rendering.md	
read	docs/en/getting-started.md	
read	docs/en/hyperparameters/data-argument.md	
read	docs/en/hyperparameters/model-argument.md	
read	docs/en/hyperparameters/sample-argument.md	
read	docs/en/hyperparameters/training-argument.md	
read	docs/en/index.rst	
read	docs/en/inference/deploy.md	
read	docs/en/installation.md	
read	docs/en/llamaboard-web-ui.md	
read	docs/en/training/dpo.md	
read	docs/en/training/sft.md	
read	docs/make.bat	
read	docs/requirements.txt	
read	docs/zh/configuration/data.md	
read	docs/zh/configuration/index.md	
read	docs/zh/configuration/inference.md	
read	docs/zh/configuration/model.md	
read	docs/zh/configuration/training.md	
read	docs/zh/developer-guide/architecture_overview.md	
read	docs/zh/developer-guide/baseplugin_mechanism.md	
read	docs/zh/developer-guide/core/base_sampler.md	
read	docs/zh/developer-guide/core/base_trainer.md	
read	docs/zh/developer-guide/core/batch_generator.md	
read	docs/zh/developer-guide/core/callback.md	
read	docs/zh/developer-guide/core/data_engine.md	
read	docs/zh/developer-guide/core/index.md	
read	docs/zh/developer-guide/core/model_engine.md	
read	docs/zh/developer-guide/core/renderer.md	
read	docs/zh/developer-guide/index.md	
read	docs/zh/developer-guide/plugins/data_plugins.md	
read	docs/zh/developer-guide/plugins/index.md	
read	docs/zh/developer-guide/plugins/kernel-acceleration/overview.md	
read	docs/zh/developer-guide/plugins/model_plugins.md	
read	docs/zh/developer-guide/plugins/trainer_plugins.md	
read	docs/zh/feature-guide/batching.md	
read	docs/zh/feature-guide/data_preparation.md	
read	docs/zh/feature-guide/distributed_training.md	
read	docs/zh/feature-guide/dpo.md	
read	docs/zh/feature-guide/index.md	
read	docs/zh/feature-guide/inference.md	
read	docs/zh/feature-guide/kernel_acceleration.md	
read	docs/zh/feature-guide/model_export.md	
read	docs/zh/feature-guide/model_saving.md	
read	docs/zh/feature-guide/optimizer.md	
read	docs/zh/feature-guide/rm.md	
read	docs/zh/feature-guide/sft.md	
read	docs/zh/index.rst	
read	docs/zh/multi-backend/npu/index.md	
read	docs/zh/quick_start.md	

Event	Path or command	Hash
read	examples/README.md	
read	examples/README_zh.md	
read	requirements/adam-mini.txt	
read	requirements/apollo.txt	
read	requirements/aqlm.txt	
read	requirements/badam.txt	
read	requirements/bitsandbytes.txt	
read	requirements/deepspeed.txt	
read	requirements/dev.txt	
read	requirements/eetq.txt	
read	requirements/fp8-te.txt	
read	requirements/fp8.txt	
read	requirements/fsdpturbo.txt	
read	requirements/galore.txt	
read	requirements/gptq.txt	
read	requirements/hqq.txt	
read	requirements/ktransformers.txt	
read	requirements/liger-kernel.txt	
read	requirements/metrics.txt	
read	requirements/minicpm-v.txt	
read	requirements/moss-vl.txt	
read	requirements/npu.txt	
read	requirements/openmind.txt	
read	requirements/sglang.txt	
read	requirements/swanlab.txt	
read	requirements/triton_ascend.txt	
read	requirements/vllm.txt	
read	requirements/xpu.txt	
read	tests/version.txt	
process	grep -vE it/s\) \$ s/it\) \$	
process	tr \r \n	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	tail -30	
process	nvidia-smi --query-gpu=count,name --format=csv,noheader	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	/var/tmp/kv-repro/028-hiyouga-LlamaFactory/env/bin/python /var/tmp/kv-rep...	
process	python - /var/tmp/kv-repro/028-hiyouga-LlamaFactory/data/run	

C Run history

Invocation 1, OK, 19 minutes 54 seconds, 75 output lines, started 2026-10-08T09:12:42

```
metric_hash  f43ffac3b06f5562cbf261050dadf162b96a2153d6bd9e1d4fbe0615b3134e0e
run_digest   aaffe8a39ffab5f27c3b57dee4e6d7fd391af4ea5e7acba2d4010cb1d74622dc
anchored     at run end
```

D Raw seal fields and hashes

Seal

```
data_hash      f8c48c4d67d199557674230f0069a6339f880f9716e54532e583e14e3bbfd2a5
signed_message_hash cd6399355d9c8b54e7d906b26a47896f74e4c5099ca24c8f0e88f9fa04773465
nonce          01f0fd55920ba6fc060a05e65b844daf
signed_at      2026-10-08T09:32:38.125000+00:00
sealed_at      2026-10-08T09:32:38.183126818Z
checkout_bundle 0a680f8869eee4c0bcad32ae50d68487a9b387ee0a0bd47537b0f6c8a9da5586
```

Signature (base64)

```
rwXveQn70Wz/vbyyzDCIc4JjBiaI3BwvfzCMzpn5Wx4BUYJaWEaI3MXGIv5rUtxkk1aFsg9o02awh2sUeH+C7Uwt
7Y2QKK1nyGUQaUyIAkI0rS82Fiyo8zWEDbxrAL60YG7HnE4wcqKLD9/0H0ML6DiXbpcfFpLAZ7V8c/RwWg147Rv7
lx+auqNDfGoqa3iuRklr992DNDvVJ3ymIhqQNH7ji9F7y9G6SpeelrWM/DFRs6hv1lU9emlz6XeJ4BKe8z2Rn1Iz
FcWYFRbn8PxrStasmYubr6XSHkWl47pfeZpu0MhTmjX0sQJ2TbCiXRGcVeJQmWtjVH8qfVuSXKu4KQeyPTUA4iF
DgF+JS0iw3ujEHrEGt4UhrRP08aqko8Y1nE89QtMycZdKLS5lbuZTwHZpIRhfgHdG3A1B4dUmHTpn6hNhhQalel0
PM+Aq0I6bCUHW4i12dCocenI/hw/qy7FxpNBjftGnmAJ4GJVqF9MBFs1lXWEw1YltXuJdPdtVcX8hnnwHIX7rdBy
WsroL+zrXnBF2r0uslBX1P/F0HoLvZqn2B1N0Gls51/55P7AZz5HTW4cskcKXqTQWpWc565RWLUsoUoX+M3zklkD
i+EmRvFP55/5ZsV0EqZUiAEc1NHFOSTXRstws8gtLFor1P4VV9FkliPpbUKwyX9yBDE=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICijANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgWP+iWscPc3HzpmjpsKbTmOemqgp6RB6GnnPpxYa5X
7qU95EfEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeAdbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNlrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlAQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syans67u7UMgIcNgeNHNCy9bLgpU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdw3gHLcCAWEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout  c99df93ad79ccc24a1081453c1bd26dbbf7c36434a245650a040aebd466c860b
stderr  e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics f43ffac3b06f5562cbf261050dadf162b96a2153d6bd9e1d4fbe0615b3134e0e
digest  aaffe8a39ffab5f27c3b57dee4e6d7fd391af4ea5e7acba2d4010cb1d74622dc
kv_run.sh (unchanged)
pre 414b1e1ace888d5b8513352135e9c2450c278e9b466cbacd554b7ef39fab4bbb
post 414b1e1ace888d5b8513352135e9c2450c278e9b466cbacd554b7ef39fab4bbb
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
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